

Exp No:3.a) Conduct an experiment to show data visualization using line plot.

Aim:

To conduct an experiment to show data visualization using line plot.

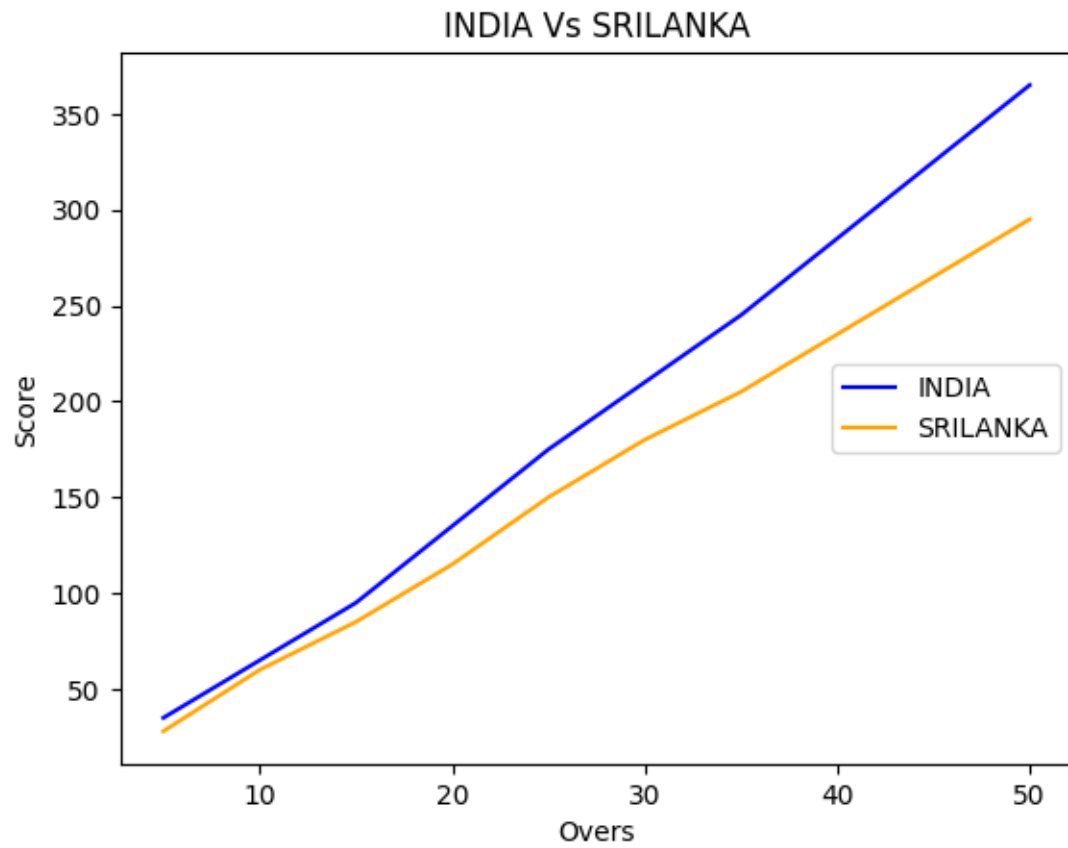
Description:

Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as cricket
Overs = list(range(5, 51, 5))
Indian_Score = [35, 65, 95, 135, 175, 210, 245, 285, 325, 365]
Srilankan_Score = [28, 60, 85, 115, 150, 180, 205, 235, 265, 295]
cricket.plot(Overs, Indian_Score, color="blue", label="INDIA")
cricket.plot(Overs, Srilankan_Score, color="orange", label="SRILANKA")
cricket.title("INDIA Vs SRILANKA")
cricket.xlabel("Overs")
cricket.ylabel("Score")
cricket.legend(loc="center right")
cricket.show()
```

output:



Result:

The line plot was successfully drawn comparing India and Sri Lanka scores over overs. India scored higher than Sri Lanka throughout the match.

Exp No:3.b) Conduct an experiment to show data visualization using bar chart.

Aim:

To conduct an experiment to show data visualization using bar chart.

Description:

Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as hscmark
import numpy as np

Names = ['ARUN', 'MEENA', 'RAJ', 'DIVYA', 'KARTHIK', 'ANU']
xaxis = np.arange(len(Names))

Percentage_hsc = [88, 76, 93, 84, 69, 97]

hscmark.bar(Names, Percentage_hsc)

hscmark.xticks(xaxis, Names, rotation=45)

hscmark.xlabel("Names of Pupil")

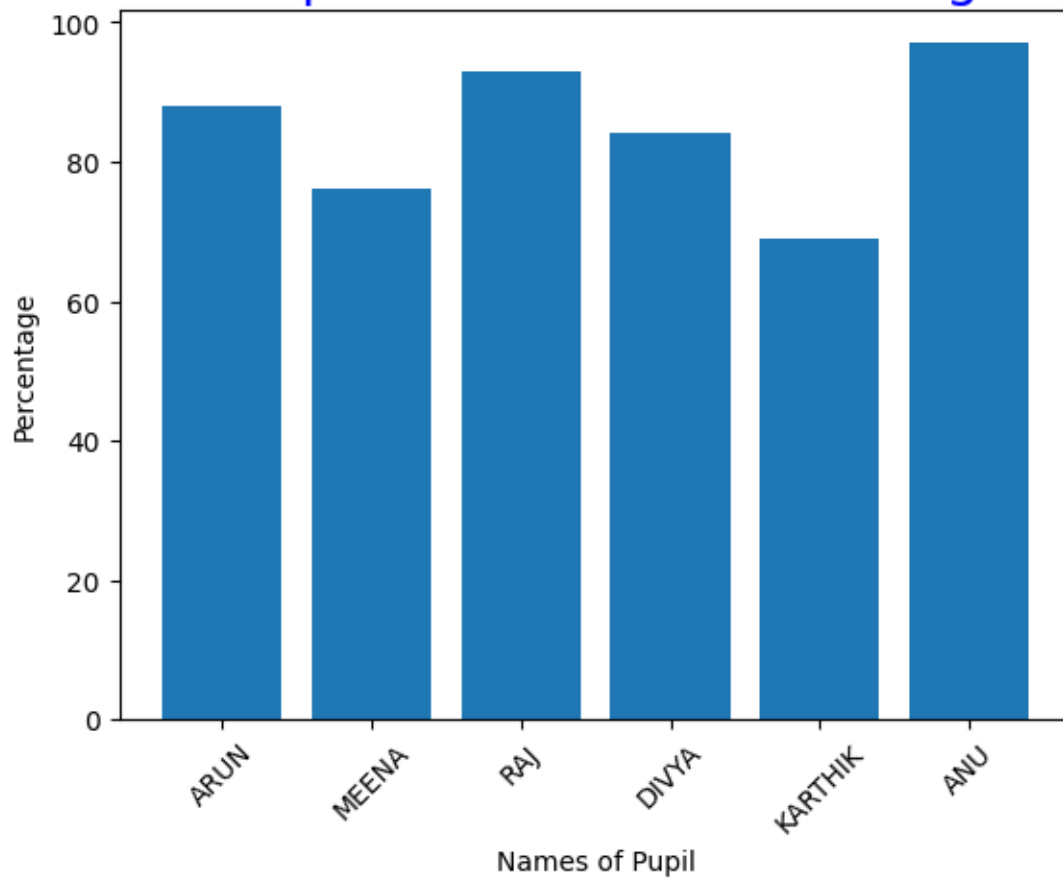
hscmark.ylabel("Percentage")

hscmark.title("Comparison of HSC Percentage", fontsize=20, color="blue")

hscmark.show()
```

Output:

Comparison of HSC Percentage



Result:

The bar chart was plotted successfully showing comparison of student's percentage. Anu scored highest (97%) and Karthick lowest (69%).

Exp No:3.c) Conduct an experiment to show data visualization using pie chart

Aim:

To Conduct an experiment to show data visualization using pie chart.

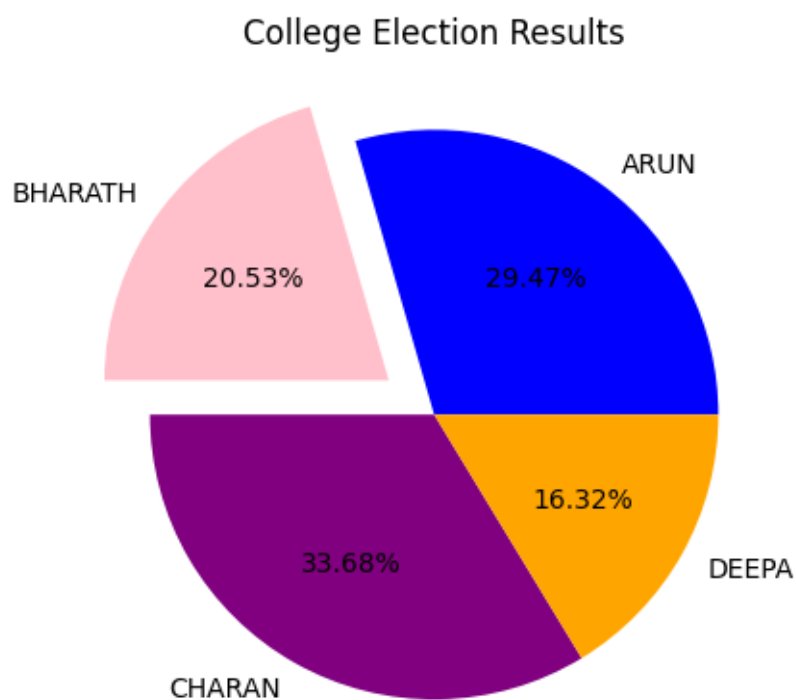
Description:

Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as election
labels = ['ARUN', 'BHARATH', 'CHARAN', 'DEEPA']
Votes = [280, 195, 320, 155]
colors = ['blue', 'pink', 'purple', 'orange']
explode = (0, 0.2, 0, 0)
election.pie(Votes, labels=labels, colors=colors,
             explode=explode, autopct='%0.2f%%')
election.title('College Election Results')
election.show()
```

Output:



Result:

The experiment was successfully completed, and the pie chart clearly represents the given data using proper labels, colors, and proportions.